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COURSE NAME:JAVA PROGRAMMING

COURSE CODE:CSA 0924

C02-AT2

Evaluate whether exposing Iterator from an internal collection preserves encapsulation. Discuss modification risks and propose safer alternatives.

Aim:

To evaluate whether exposing an **Iterator** from an internal collection preserves encapsulation and to demonstrate a safer approach using an **unmodifiable iterator**.

Pseudocode:

START

Create a list with some elements

Return an unmodifiable iterator

Display all elements using iterator

Try to modify the collection

IF modification is attempted THEN

 Display "Modification not allowed"

END IF

STOP

Java Code:

```
import java.util.*;

class StudentList {
    private List<String> students = new ArrayList<>();

    StudentList() {
        students.add("Alice");
        students.add("Bob");
        students.add("Charlie");
    }

    public Iterator<String> getIterator() {
        return Collections.unmodifiableList(students).iterator();
    }
}

public class Main {
    public static void main(String[] args) {
        StudentList obj = new StudentList();
        Iterator<String> it = obj.getIterator();
        while (it.hasNext()) {
            System.out.println(it.next());
        }
        System.out.println("Collection is protected from modification.");
    }
}
```

Input:

Alice

Bob

Charlie

Output:

Alice

Bob

Charlie

Collection is protected from modification.


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Main.java



Run

```
1 import java.util.*;
2
3 class StudentList {
4     private List<String> students = new ArrayList<>();
5
6     StudentList() {
7         students.add("Alice");
8         students.add("Bob");
9         students.add("Charlie");
10    }
11
12    public Iterator<String> getIterator() {
13        return Collections.unmodifiableList(students)
14            .iterator();
15    }
16
17    public class Main {
18        public static void main(String[] args) {
19
20            StudentList obj = new StudentList();
21
22            Iterator<String> it = obj.getIterator();
23
24            while (it.hasNext()) {
25                System.out.println(it.next());
26            }
27        }
28    }
29 }
```

Output

Alice
Bob
Charlie
Collection is protected from modification.

=== Code Execution Successful ===



Main.java	Output
<pre>1 ~ 2 3 ~ 4 ~ 5 ~ 6 ~ StudentList() { 7 students.add("Alice"); 8 students.add("Bob"); 9 students.add("Charlie"); 10 } 11 12 ~ public Iterator<String> getIterator() { 13 return Collections.unmodifiableList(students 14).iterator(); 15 } 16 17 ~ public class Main { 18 public static void main(String[] args) { 19 20 StudentList obj = new StudentList(); 21 22 Iterator<String> it = obj.getIterator(); 23 24 ~ while (it.hasNext()) { 25 System.out.println(it.next()); 26 } 27 28 System.out.println("Collection is protected 29 from modification."); 30 } 31 }</pre>	<pre>Alice Bob Charlie Collection is protected from modification. === Code Execution Successful ===</pre>